Chapter: 23

DIGITAL ELECTRONICS

INTRODUCTION:

Definition:

Digital electronics is a branch of electronics that deals with the manipulation of digital signals, represented as binary digits (0s and 1s).

Explanation:

Digital electronics is a branch of electronics that works with **numbers instead of smooth signals**.

In digital electronics, information is written using only **two values**:

- **0** → OFF (no current / low signal)
- **1** → ON (current present / high signal)

These 0s and 1s are called **binary digits**.

Unlike analog electronics, which uses continuous signals, digital electronics processes discrete signals, enabling precise control and storage of information.

This field encompasses the design, analysis, and application of digital circuits and systems, forming the foundation for modern computing, communication, and information processing technologies.

DIGITAL SIGNAL LEVELS:

In digital electronics, signal levels are represented in various ways. Here are common digital signal levels:

1. LOW LEVEL (0):

- This represents the binary digit 0. In terms of voltage, it is associated with a lower voltage level.
- Often referred to as "low" or "logic 0."

2. HIGH LEVEL (1):

- This represents the binary digit 1. In terms of voltage, it is associated with a higher voltage level.
- Often referred to as "high" or "logic 1".

3. THRESHOLD LEVEL:

The threshold level is the voltage level that separates low and high states.

- Signals below this threshold are interpreted as 0, and signals above it are interpreted as 1.
- The threshold level helps define the noise margin and ensure reliable signal interpretation.

4. LOGIC LEVELS:

- Depending on the technology and the system, different logic levels might be used.

5. SWING OR VOLTAGE RANGE:

The difference in voltage levels between low and high states is often referred to as the swing or voltage range.

For example, voltage varies from -5V to +5V will have a swing or voltage range of 10 V.

A larger voltage swing can enhance noise immunity and signal reliability.

UNDERSTANDING DIGITAL SIGNAL LEVELS:

In digital electronics, the 'high' and 'low' states are often symbolically represented as '1' and '0' respectively. This binary notation helps to describe the states of open and closed circuits.

HIGH STATE (1):

When a digital signal is in a 'high' state, it is typically associated with a closed circuit or a voltage level close to the maximum specified value.

The symbolic representation for this state is “1”. This state signifies the “ON” or “active” state in many digital systems.

LOW STATE (0):

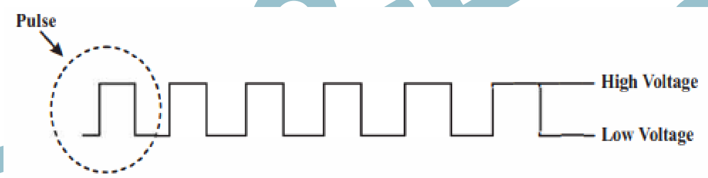
Conversely, the 'low' state is linked to an open circuit or a voltage level close to the minimum specified value.

It is symbolically represented as “O”. This state signifies the “OFF” or “inactive” state in digital systems.

Explanation:

High and Low represent the signal levels in the circuit. Digital waveforms consist of voltage levels that are changing back and forth between the HIGH and LOW states.

These can also be described as the one that is composed of a series of pulses, or a pulse train as shown in Figure.



These symbolic representations simplify the expression and interpretation of digital information, allowing for clear communication and logical operations within electronic circuits.

The '1' and '0' states form the basic building blocks for encoding and processing binary data in digital systems.

This binary representation forms the fundamental language of digital systems, allowing them to process and manipulate data efficiently

LOGIC GATES:

Definition:

Logic gates are the basic components of the digital circuit with one output and more than one input.

Logic gates are tiny electronic switches that enable decision-making in computers and other devices.

They combine to perform logical operations, playing a crucial role in information processing.

TYPES OF LOGIC GATES:

- AND, OR, and NOT gates are the basic gates, while
- NAND and NOR are the universal gates.
- EX-OR and EX-NOR are the special gates.

1. AND GATE:

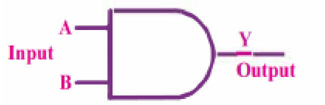
The AND gate produces a 'high' output only when all its inputs are “high”.

In other words:

The output of an AND gate attains state 1 if and only if all the inputs are in state I.

Boolean expression of AND Gate:

1. Two-input Y = AB
2. Three-input Y = A.B.C
3. Four input Y = A.B.C.D



Truth Table of AND:

Two Inputs; Y= A. B			Three Inputs; Y = A. B. C			
A	B	Y	A	B	C	Y
0	0	0	0	0	0	0
0	1	0	0	0	1	0
1	0	0	0	1	0	0
1	1	1	0	1	1	0
			1	0	0	0
			1	0	1	0
			1	1	0	0
			1	1	1	1

2. OR GATE:

An OR gate generates a "high" output if any of its inputs are 'high'.

In other words:

The output of an OR gate attains state 1 if one or more inputs attain state 1.

Boolean expression of OR gate:

1. Two-input Y = A+B
2. Three-input Y = A+B+C
3. Four input Y = A+B+C+D

Truth Table of OR gate:

Two Input; Y = A + B			Three Input; Y = A + B + C			
A	**B**	**Y**	**A**	**B**	**C**	**Y**
0	0	1	0	0	0	0
0	1	1	0	0	1	1
1	0	1	0	1	0	1
1	1	1	0	1	1	1
			1	0	0	1
			1	0	1	1
			1	1	0	1
			1	1	1	1

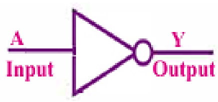
3. NOT GATE:

It is used to perform the inversion operation in digital circuits.

For example: the output of a NOT gate attains state 1 if and only if the input does not attain state 1.

Boolean expression of NOT gate:

The Boolean expression is y = $\bar{A}$, It is read as Y equals NOT A.



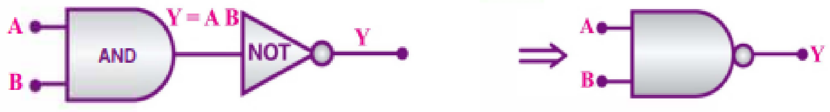
Truth Table of NOT gate:

The truth table of NOT gate is given in table.

Truth Table No. 23.3 NOT GATE	
A	**Y**
0	1
1	0

4. NAND GATE:

A NAND gate is an AND gate followed by a NOT gate, yielding the opposite output of an AND gate.



The Boolean expression of NAND gate:

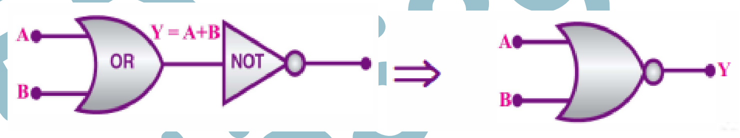
1. Two-input; Y = $\overline{A.B}$
2. Three inputs; Y = $\overline{A.B}$
3. Four inputs; Y = A.B.C.D

The Truth Table:

Two Input; Y=$\overline{A.B}$			Three Inputs; Y=$\overline{A.B.C}$			
A	B	Y	A	B	C	Y
0	0	1	0	0	0	1
0	1	1	0	0	1	1
1	0	1	0	1	0	1
1	1	0	0	1	1	1
			1	0	0	1
			1	0	1	1
			1	1	0	1
			1	1	1	0

5. NOR GATE:

A NOR gate is an OR gate followed by a NOT gate, providing the opposite result of an OR gate.



The Boolean expression of NOR gate:

1. Two-input Y=A+B
2. Three-input y =A+ B + C
3. Four input y = A + B + C + D

The Truth Table:

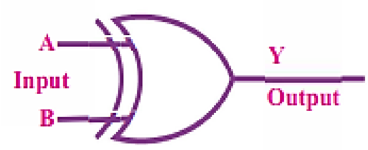
Two Input; Y = $\overline{A+B}$			Three Inputs; Y = $\overline{A+B+C}$			
A	B	Y	A	B	C	Y
0	0	1	0	0	0	1
0	1	0	0	0	1	0
1	0	0	0	1	0	0
1	1	0	0	1	1	0
			1	0	0	0
			1	0	1	0
			1	1	0	0
			1	1	1	0

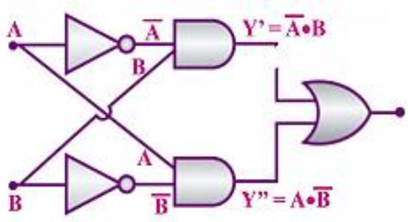
6. XOR GATE (EXCLUSIVE-OR GATE):

The XOR gate outputs a 'high' signal if the number of 'high' inputs is odd.

In other words:

The output of a two-input XOR gate attains state 1 if one adds only input and attains state 1.





The Boolean expression of XOR gate:

1. Y = A.$\overline{B}$ + $\overline{A}$
2. Three-inputs; Y = A.$\overline{B}$.$\overline{C}$ + $\overline{A}$.B.$\overline{C}$ + $\overline{A}$.$\overline{B}$.C + A.B.C
3. Four inputs; Y = A$\oplus$ B$\oplus$ C$\oplus$ D

Truth Table of XOR gate:

Two Inputs; Y = A. $\overline{B}$ + $\overline{A}$			**Three Inputs: Y = A.$\overline{B}$.$\overline{C}$ + $\overline{A}$.B.$\overline{C}$ + $\overline{A}$.$\overline{B}$.C + A.B.C**			
A	B	Y	A	B	C	Y
0	0	1	0	0	0	1
0	1	0	0	0	1	0
1	0	0	0	1	0	0
1	1	0	0	1	1	0
			1	0	0	0
			1	0	1	0
			1	1	0	0
			1	1	1	0

7. XNOR GATE (EXCLUSIVE-NOR GATE):

In the XNOR gate, the output is in state 1 when both inputs are the same, that is, both O or both 1.

The Boolean expression of XNOR gate:

1. Two inputs; Y = (($A\oplus\overline{B}$)) = (A.B + $\overline{A}$. $\overline{B}$)
2. Three-inputs; Y = (($\overline{A}$). ($\overline{B}$) $\overline{C}$) + (A.B.C)
3. Four –inputs; Y = ($\overline{A}$. ($\overline{B}$.) $\overline{C}$. $\overline{D}$) + (A.B.C.D)

Truth Table of XNOR gate:

Two Inputs **Y = (($A\oplus\overline{B}$)) = (A.B + $\overline{A}$. $\overline{B}$)**			**Three Inputs** **Y = (($\overline{A}$). ($\overline{B}$) $\overline{C}$) + (A.B.C)**			
A	B	Y	A	B	C	Y
0	0	1	0	0	0	1
0	1	0	0	0	1	0
1	0	0	0	1	0	0
1	1	1	0	1	1	1
			1	0	0	0
			1	0	1	1
			1	1	0	1
			1	1	1	0

DEMONSTRATION OF LOGIC GATES IN SIMPLE CIRCUITS:

In this part, we are going to have some fun with simple circuits. By utilizing just 2 switches, a lamp and a battery, we explore the behaviors of a 2-input AND gate and a 2-input OR gate.

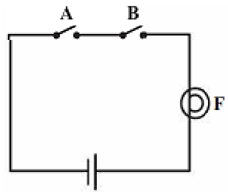
These hands-on examples offer a concrete insight into how logic gates process signals and make logical decisions, forming the basis for more complex digital systems.

1. AND GATE DEMONSTRATION:

Components:

In this demonstration, we will use the following components as shown in the circuit diagram:

- 2 switches (Switch A and Switch B)
- 1 lamp
- 1 battery



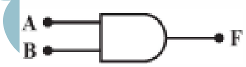
The presented circuit exhibits two switches (A & B), and a bulb (F) arranged in series with a cell.

According to basic circuit theory, the bulb illuminates only when a closed circuit permits the flow of current.

Behavior:

The lamp will light up only when both Switch A and Switch B are in the ON position. If either or both switches are OFF, the lamp remains off.

However, in the context of digital electronics, we approach the analysis differently. Instead of focusing on the physical flow of current, we interpret the circuit as making decisions, determining whether the output represents logic 1 (the bulb is lit) or logic 0 (bulb not lit). This decision is contingent upon the logical states of the inputs (i.e., the switches), where an open switch corresponds to logic 0 and a closed switch corresponds to logic 1.



A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

The truth table and the equivalent combination of above circuit in logic gate is shown in figure.

2. OR GATE DEMONSTRATION:

Components:

In this demonstration, we will use the following components as shown in the circuit diagram:

- 2 switches (Switch A and Switch B)
- 1 lamp
- 1 battery

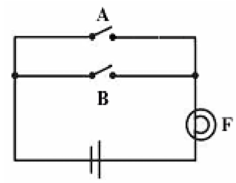


Figure shows a simple circuit which contains two switches in parallel.

Behavior:

The lamp will light up if either or both Switch A and Switch B are in the ON position. The lamp remains off only when both switches are OFF.

The truth table and the equivalent combination of above circuit in logic gate is shown in figure.

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

In these circuits:

- For the AND gate, the output is HIGH (lamp on) only when both inputs are HIGH.
- For the OR gate, the output is HIGH (lamp on) when at least one input is HIGH.

These simple circuits demonstrate the basic principles of “AND” and “OR” gates, showcasing how they process signals based on logical operations.

SUMMARY

- **Digital electronics** deals with digital signals represented by binary digits (Os and 1s).
- **Digital signals** use discrete voltage or current levels to represent information.
- **Common digital signal levels** include high level (logic 1), low level (logic 0), threshold level, and voltage swing.
- **Logic gates** are fundamental building blocks of digital circuits that perform logical operations on binary data.
- **Basic logic gates** include AND, OR, and NOT gates
- **NAND and NOR gates** are universal gates that can be used to create any other logic gate.
- **XOR and XNOR gates** are special logic gates used for specific operations.
- **Logic gates can be implemented using transistors** which act as electronic switches.
- **Simple circuits with switches and bulbs** can be used to demonstrate the behavior of AND OR gates.

TEXTBOOK WORK

SECTION- A: MULTIPLE CHOICE QUESTIONS (MCQS)

1. The voltage level, typically associated with a 'low' state in digital electronics is:
   (a) -5 volts (b) 0 volts (c) +5 volts (d) +10 volts
2. In binary representation, the corresponding '0' in digital electronics is: **(2024)**
   (a) Low state (b) High state (c) Open circuit (d) Closed circuit
3. The primary purpose of a Truth table in digital electronics is to: **(2025)**
   (a) Determine input voltage (b) Measure circuit resistance
   (c) Analyze output states based on input combinations (d) Calculate circuit power
4. Which logic gate combination produces an output of 'I' when at least one input is '1'?
   (a) AND (b) OR (c) XOR (d) NOT
5. In a circuit with a 2-input OR gate, the status of the lamp if either switch is closed
   (a) OFF (b) ON (c) Blinking (d) Flickering
6. The behavior of 2 input AND gate, when both switches are closed, Lamp is
   (a) OFF (b) ON (c) blinking (d) flickering
7. In a 3-input OR gate, if all inputs are '0'so the output will be: **(MP:25)**
   (a) 1 (b) 0 (c) Undefined (d) Both A and B
8. If the input is '1', The output of a NOT gate will be:
   (a) 1 (b) 0 (c) Undefined (d) Both A and B
9. In a 2-input XOR gate, when both inputs are '1' then the output will be
   (a) 1 (b) 0 (c) Undefined (d) Both A and B
10. Which combination of logic gates is equivalent to an XOR gate?
    (a) AND+ OR (b) NOT+ NOR (c) OR+ NANO (d) AND+ NANO

SECTION- B: SHORT ANSWERED QUESTIONS

1. Explain the significance of signal levels in digital electronics and how they are represented in terms of voltage. **(2024)**
2. Describe the binary representation used in digital electronics. How is this representation related to the 'high' and 'low' states?
3. Choose any three logic gate symbols and explain their operations.

4. Create a combination of logic gates that mimics the behavior of an XOR gate.

SECTION- C: LONG ANSWERED QUESTIONS

1. Design a simple circuit using a 2-input AND gate, a switch, a lamp, and a battery. Explain how the circuit operates and the conditions under which the lamp will be illuminated.
2. Construct a truth table for a 3-input OR gate. Explain how the truth table represents the relationship between input combinations and the resulting output.

PAST YEAR PAPERS:

1. Draw the circuit symbol of OR and AND gates and write Truth table for two inputs. **(2025)**
2. What are logic gates? Describe the binary operation of OR gate or AND gate for three inputs. **(MP:25)**